James Cenawood

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EDUCATION

Cornell University, College of Engineering, Ithaca, NY — Aug 2024 - May 2028

Bachelor of Science in Computer Science, GPA: 3.9

Relevant Courses: Object-Oriented Programming & Data Structures, Functional Programming, Probability & Statistics (A+), Stochastic Processes (A+), Linear Algebra; Machine Learning, Simulation Modeling & Analysis, and Analysis of Algorithms in progress

EXPERIENCE

Northwell Health | Data Science Intern — *New Hyde Park*, NY — Jun 2026 - Aug 2026

- Built an NL-to-SQL agent (Python, LangGraph) with deterministic schema checks and **no patient-data access**
- Built a stratified evaluation benchmark across 74 answerable questions that attributed **82.4% of top-5 retrieval misses to vocabulary mismatch** rather than ranking depth
- Closed the gap by fusing BM25-ranked SQLite FTS and dense-vector retrieval over 40,551 Epic schema files and 482,130 indexed chunks, raising hit@5 from **53.8% to 75% across ~250 analyst queries**

Cornell Physical Intelligence | Software Subteam Lead — *Ithaca*, NY — Dec 2025 - Present

- Led an **eight-engineer subteam**, owning the research agenda and implementation for mapless visual navigation and collective-thrust/body-rate control in Anduril's AI Grand Prix; **passed Virtual Qualifier 1 at 17.2s**
- Prevented invalid policies from reading unavailable race-time state and automated simulator launch, scoring, telemetry and visual capture, and summaries with an experiment harness (Python, PowerShell) for coding agents
- Designed and administered **20+ technical interviews** using an original drone-themed algorithms problem set

PROJECTS

Surpassing the Leader | Exact Stochastic-Game Solver (GitHub) — 2026

- Formalized a finite simultaneous-move zero-sum stochastic game; implemented the game, solver, and verification pipeline (Python, OCaml, C++, Rust) and authored the paper
- Proved a value-preserving quotient reducing **5.27 billion reachable states to 289.4 million equivalence classes (18.2×)** and a strictly increasing potential ordering them into 1,201 layers, eliminating game-tree search
- Solved the full 289-million-state Drop the Handkerchief game by certified backward induction **in 50 seconds, versus a projected ~5 years**; the first mover wins 54.5% of the time under optimal play

AlgoSplit | Quantitative Hypertrophy Modeling (algo-split.vercel.app) — 2025 - 2026

- Shipped authenticated web and iOS apps (FastAPI, Supabase, React, Expo) for a 29-muscle-region training research engine with workout logging, split comparison, and 3D stimulus visualization
- Derived per-set marginal stimulus curves from hypertrophy research, combining RIR, load, and reps into one set score; optimizing that objective raised net weekly stimulus **35% on average across 100+ user-submitted splits**
- Cut uncached p95 latency from **31.892 ms to 3.847 ms (8.29×)** by porting the stimulus backtesting and analysis kernel to Rust with Python parity within 1×10^{-8}

Sentinel | Intelligent Waiting-Room Triage, **1st Place**, URMC x WICC x Gemini Hackathon (Demo) — 2026

- Built a QR-based waiting-room triage platform (TypeScript, LangChain, Gemini API) whose deterministic engine converts conversational intake, symptoms, vitals, and wait time into **ESI estimates, risk scores, follow-up actions, and a live queue**
- Designed a safety-controlled queue with emergency overrides, confidence/noise penalties, promotion hysteresis, and delayed demotion; constrained the Gemini reviewer to within-band reordering, escalations, and follow-ups

SKILLS & HONORS

Languages: Python, OCaml, C++, SQL, Rust, TypeScript

Tools & Frameworks: PyTorch, scikit-learn, NumPy, pandas, FastAPI, LangGraph, React, PostgreSQL, Docker, Git

Honors: 1st Place, URMC x WICC x Gemini Hackathon (2026); Tied 1st, Northwell Intern Kaggle Competition (2026); Cornell Poker Club: 3rd, Citadel $2K Guaranteed; 3rd, Jane Street Freeroll

Interests: Tournament Poker, Game Design, 3D Animation, Rock Climbing